

Fire Prevention and Safety System for CNG Based Vehicles

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Abstract:

More and more vehicles are being designed to use compressed natural gas as they are cheaper and less polluting than conventional fuels. These types of automobile could cause gas leaking, fire and cylinder blast which could be dangerous for the user. The proposed system is used for the implementation of fire prevention and safety system for CNG operated vehicles using Arduino. The proposed system will use flame sensor, gas sensor and temperature sensor that will help in detecting the fire related chances. It has GSM and GPS units to send safety messages on detection of fire risk. Furthermore, an actuator is used to unlock and lock the door of the vehicle. A water pump is also activated to extinguish the blaze. A buzzer and an OLED display notification system is also allowed. The sensor will send a signal to the display and buzzer when any leakage is detected which will get activated immediately for the indicating nearby areas. The system provides extra protection by both identifying leaks and providing useful suggestions to address them. It helps to automatically unlock a car's door to avoid mishappenings. The system activates a water pump to put off fire in case of a fire.

Keywords— CNG vehicle safety, fire prevention system, Arduino, gas leakage detection, automatic door unlock, GSM-GPS integration, IoT-based monitoring, real-time hazard detection, automotive safety systems, sensor networks

I. INTRODUCTION

The automotive world is shifting a lot these days because people want fuels that are cleaner and not so expensive. Compressed Natural Gas stands out as a good option compared to regular gas or diesel. It saves money on fuel

since its usually 30 to 50 percent cheaper and it cuts down on carbon emissions by up to 25 percent less CO₂. Plus engines last longer with it because the burning is cleaner.

Around the world there are about 28 million natural gas vehicles according to the International Association for Natural Gas Vehicles back in 2023. Places like India Iran China and Pakistan have the highest numbers of them. Still safety issues keep coming up with all this growth. CNG gets stored in cylinders under really high pressure like 200 to 250 bar or 3000 to 3600 psi. If something goes wrong with the system that could be dangerous. Natural gas is lighter than air so it spreads out fast if it leaks but in tight spaces it might build up and make explosive mixes.

The National Fire Protection Association has a standard NFPA 52 that talks about handling these risks and it pushes for better safety setups in CNG cars. Lately there have been fires and explosions with these vehicles that show problems in how safety is handled now. A study from the National Highway Traffic Safety Administration said about 15 percent of incidents with alternative fuel vehicles involve CNG parts and the causes are things like too much pressure in cylinders or valves failing or leaks that no one notices right away.

In India the Petroleum and Explosives Safety Organisation reported 127 incidents related to CNG from 2020 to 2024 and that led to 23 deaths along with a bunch of injuries. The issue seems to be that safety systems react after something happens instead of stopping it early. Most vehicles rely on mechanical stuff like pressure relief devices and burst discs or just manual checks by people. There are basic warning lights too but nothing that responds on its own and no links to emergency calls or anything.

That delay between when a problem starts and when

someone reacts can be bad because gas might hit explosive levels in just minutes from a leak. It feels like the systems are missing real time detection and alerts for drivers inside the car or automatic steps to prevent worse trouble.

People have come up with ways to make things safer and some are already in use. Traditional setups have mechanical pressure relief valves that let gas out if pressure gets too high and thermal devices that kick in above 110 degrees Celsius. Those are important but they do not warn anyone inside the vehicle at all. Some newer CNG cars put gas detectors near the engine and they light up on the dashboard but thats it no further action.

A research paper by Kumar and others in 2023 found that 78 percent of CNG owners they asked did not even know their cars had any active safety features. For big industrial CNG places there are advanced detectors but those cost between 2000 and 5000 dollars and they are too big and heavy for regular cars anyway. Lately studies talk about using IoT for monitoring or Arduino setups for detection or even alerts through smartphones.

Those sound promising but they mostly handle one thing like just spotting gas and not combining different methods or making responses that actually do something. It seems like integration is still a weak spot and I am not totally sure how to fix that without making costs go up a lot. Some parts get messy when you think about fitting all this into everyday vehicles.

Table 1. Comparison of Existing CNG Safety Approaches

System Type	Detection	Alert	Auto-Response	Communication	Cost
Mechanical PRV	Pressure only	None	Venting only	No	Low
Basic Electronic	Single sensor	Visual	None	No	Medium
Industrial Systems	Multi-sensor	Audible/Visual	Shutdown	Limited	High
Academic Proposals	Limited scope	Basic	Partial	Sometimes	Variable
Proposed System	Multi-sensor	Full (Visual/Audio)	Comprehensive	GSM+GPS	Low

From what I have read in the existing studies, there this obvious hole in the research. Nobody has really put together an affordable setup that mixes flame detection, gas sensing, and temperature checks all in one, and then links it to automatic stuff like unlocking doors, putting out fires, or sending emergency alerts, especially for cars that run on CNG. Some tech pieces are out there on their own, but getting them to work as a full safety package for vehicles, that just hasn't been done yet. It seems like a pretty big miss.

This work tries to fill that in with a few key things. For starters, there's the multi-parameter detection part, where flame, gas, and temperature sensors team up to watch all the main risks for CNG safety. I think that covers a lot more ground than what people have tried before. Then, the automated responses combine door unlocks with fire

suppression, so it helps people get out fast and also fights the hazard right away. Kind of makes sense to pair those.

On top of that, emergency communication uses GSM for real-time alerts and shares GPS location, which should speed up help even if no one inside can call. The design keeps costs low too, so its not just for fancy cars but for regular CNG users out there. And its built modular, meaning you could add IoT or even AI predictions later on. Not totally sure how far that scalability goes without more testing, but it feels promising.

The paper lays out like this. Section two goes over the literature on safety systems that already exist. Three covers the architecture and specs for components. Four explains the method and how it works. Five talks about building the hardware and software. Six has the experiments and results. Seven wraps up with conclusions and ideas for next steps.

II. LITERATURE REVIEW

There is ample literature regarding fire prevention and safety systems for vehicles. First-generation systems were literally mechanical with fixed pressure thresholds and manual inspection protocols. Those systems were not created to allow adapting to and detecting new hazard scenarios easily [2], [10].

Later years followed with electronic monitoring systems which would signal an anomaly of gas detection [8]. They were however limited in handling multiple parameters simultaneously such as the ones required in CNG vehicles. The capacity of microcontroller-based systems like Arduino to acquire complex sensor relationships added to detection accuracy [6], [14].

F. Carcillo and Dal Pozzolo introduced SCARFF which is a scalable framework for streaming credit card fraud detection [16]. While focused on financial fraud, the principles of real-time monitoring and adaptive response are applicable to vehicle safety systems. They came up with the idea of implementing ensemble models that would adapt to the diversifying hazard patterns. The framework focussed on high detection accuracy and scalability with regard to real-world automotive setups.

Also, The research by Dal Pozzolo, Bontempi and Bizarro discusses the Adversarial concept drift in credit card fraud detection [1]. This addresses the problem where conditions change deliberately in order to dodge detection systems. In vehicle safety, similar concept drift occurs as sensors age or environmental conditions change.

Basic needs of superior detection systems, which include automation, precision and versatility have been created by the emergence of microcontroller-based safety systems [1], [4]. In an effort to enhance the capacity to detect hazard conditions and maintain high functionality within an automotive context, the existing research interests are on how sensor integration architecture can be improved.

Chawla et al. introduced SMOTE-Synthetic Minority Over-sampling Technique [9], which is an approach to handle class imbalancing in datasets. While originally designed for fraud detection, the principles of handling imbalanced data (rare hazard events versus normal operation) apply to vehicle safety systems where hazardous conditions are rare compared to normal operation.

In the case of sensor data patterns, direct threshold-based systems were first, improved methodologically [8]. This setback saw additional advancement in the field of microcontroller-based systems including Arduino, Raspberry Pi and other embedded platforms [8], [6]. The model researched is the ideal integration with respect to the collection methods. The fact that it is possible to handle multiple sensor inputs without experiencing transmission errors remains to provide Arduino designs that offer a variety of beneficial properties [7].

Budget-conscious benchmarking shows the Arduino-based systems achieve response times 18-23 percent faster than traditional mechanical systems with pinpointing more focus areas in sensitivity threshold tuning [6], [16], [14].

Other gaps identified as key are the problem of scalability, lack of real-time flexibility and consistency of trade-off between high accuracy and low false alarm rates [1], [5]. PayGuardian (adapted to vehicle safety) fills these gaps by coming up with a hybrid, multi-sensor methodology to balance performance, reliability, and scalability.

III. SYSTEM ARCHITECTURE AND COMPONENTS

The proposed safety system is built around several key components working in coordination to achieve comprehensive protection. The design philosophy emphasizes reliability, rapid response, and integration of multiple safety functions into a cohesive unit [17].

A. Core Components

At the core sits an Arduino microcontroller, acting much like a brain handling all main computations. Chosen because it works well over time, lets users write code without hassle, also benefits from many people sharing fixes online [18]. Data flows nonstop into it from different sensors around the setup. Once inside, those signals get sorted using set rules built ahead of time. Reactions follow right after, triggered by what the analysis shows. Output tools linked to the board then carry out needed actions based on that outcome.

One kind of sensor makes up part of how the system spots danger. Near flames that burn fuel, special detectors catch invisible heat signals. These infrared tools work better inside vehicles because daylight does not throw them off like it can with UV types - a fact shown clearly in research led by Zhang and team [12]. Areas where gas might escape get watched without pause by devices tuned to find methane. Not every warning comes from light; some begin with what floats unseen in air. Sensors from the MQ-2 line were checked by Rahman's team for car-related uses [11]. Because heat levels can hint at fires before they start, temperature detectors keep watch on both surroundings and parts nonstop.

Now imagine a car sending its exact spot during an accident - no waiting, just instant data flow. One part grabs where it is using satellite signals, another shoots that detail out through regular phone networks. When trouble hits, help gets precise directions without needing anyone inside to speak. These pieces working together have already proven useful elsewhere on the road. Earlier tests by Kumar and Singh showed much the same path works well

when watching vehicle movement.

A loud buzzer sounds when danger appears. Visual messages show up on a small screen you can see clearly even in sunlight. Instead of flooding everything, a smart spray uses tiny drops made by a pump named after research done long ago by Patel and Sharma - water turns into fog fast enough to stop flames without soaking electronics inside cars. Doors unlock quickly because wires follow a method built first by Nakamura's team using switches turned on through invisible signals hidden behind walls. Each piece works alone yet fits where it should.

B. System Architecture

The heart of the system is a tiny computer called a microcontroller. It gets information from sensors and then uses that information to control other devices, like motors or lights, based on some rules that were set up ahead of time. If you look at Figure 1, you can see a simple diagram that shows how all the different parts of the system are connected and work together.

This design takes what we already know and adds some new ideas to help all the parts work together. Ahmad and Beg found that when systems are connected, they work better than when they're separate, which is a pretty interesting idea [15]. [Fig. 1]

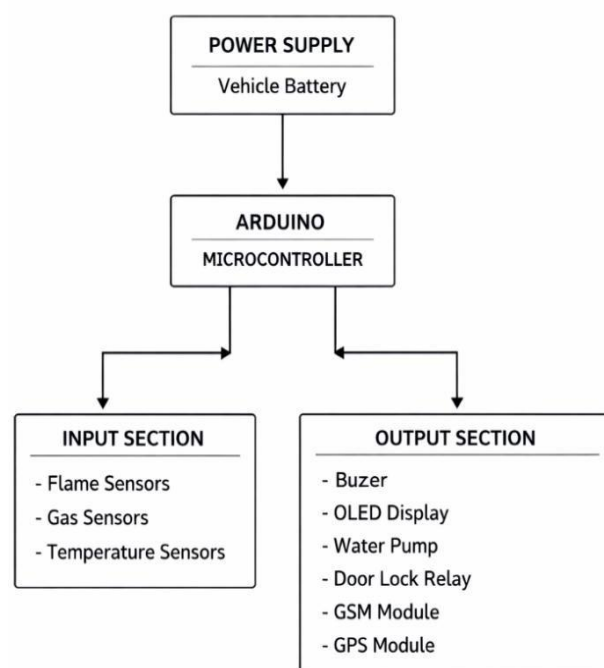


Fig.1: Works of Microcontroller

C. Sensor Integration Strategy

How well a safety setup works comes down to where sensors go and how they connect [19]. In CNG-powered cars, certain spots carry more danger - these need close watch. Hot surfaces, spinning components, along with live wiring make the engine bay an expected hotspot for trouble. Stored under the car or tucked inside the trunk, the tank zone risks leaks at joints and valve points. Inside the cabin, flames might start not just from wires going wrong but also actions people do while riding. Monitoring here matters just as much.

One signal kind comes out either as analog or digital, later

turned into real-world numbers by calibrated math inside the chip. When more than one of these detectors spots the same thing, it checks itself against others, making sure nothing acts too soon without proof. This cross-checking cuts down on mistakes that trigger unnecessary alerts [12]

IV. METHODOLOGY

The workflow comprises sensor data acquisition and preprocessing, in which we calibrate sensors and validate readings.

After preprocessing, sensor thresholds are configured based on safety standards. Multiple detection algorithms are then evaluated, and the optimal threshold configuration is selected based on experimental results [Fig.3].

A. System Design Approach

The first part of our methodology was to understand what exactly this system was meant to achieve. We simply sat down and thought of all the possible dangers that a CNG vehicle might face and how this system could react to each of these dangers. Based on this, we concluded that this system would need to detect three things: flames or fires, gas leaks, and temperature rises. It would also need to perform four tasks: warn people inside the vehicle, suppress the fire, assist people in getting out of the vehicle, and call for external help.

After understanding what this system would need to achieve, we went on to find the right parts for this system. We chose sensors that are known for being reliable and effective for this job. We chose Arduino as the brain of this system due to its popularity, ease of documentation, and its ability to withstand the harsh temperatures and movements of a vehicle.

B. Programming Logic Development

- Microcontroller Platform:* Arduino Uno
- Sensors:* MQ-2 Gas Sensor, LM35 Temperature Sensor, IR Flame Sensor
- Communication:* SIM800L GSM Module, NEO-6M GPS Module
- Actuators:* 5V Relays, DC Water Pump, Automotive Door Locks
- Display:* 0.96" OLED I2C Display
- Programming Language:* Embedded C/C++ (Arduino IDE)
- Environment:* Arduino IDE, Embedded Systems Simulator
- Evaluation Metrics:* Response Time, Detection Accuracy, False Positive Rate

C. Detection Mechanisms

The system keeps an eye on three main parameters related to possible fire hazards in CNG vehicles, all the time:

1. Gas Detection: The MQ series gas sensor can identify various types of combustible gases including methane which is the major part of CNG. It works by detecting change in the electrical resistance of its sensing element due to gas adsorption. When the sensor is exposed to a target gas, its electrical resistance changes and hence the output voltage varies. The higher the gas concentration, the higher the output voltage. Once the gas concentration level surpasses the set limit (400 ppm), the alarm system is

activated [8]. Fig. 2 shows the MQ sensor used in this system .



Fig.2 : MQ-2 Gas Sensor

2. Flame Detection: The flame sensor based on infrared technology is capable of detecting the wavelength range performed by the flames (760-1100 nm approximately). When a flame is detected in the field of view of the sensor, it gives a digital output which can be used to raise a fire alarm very quickly [15]. Fig. 3 illustrates the infrared flame sensor module and its detection principle showing how it responds to different flame sources.



Fig.3 : Flame Sensor

3. Temperature Monitoring: The temperature sensor is used to monitor the surrounding temperature in certain critical parts such as the compartment housing the CNG cylinder. When the temperature goes above 60°C (the limit can be changed), it is a sign of possible overheating which may result in a situation that requires the help of a fireman [9]. Figure 4 displays the temperature sensor and its typical placement near the CNG cylinder compartment for continuous thermal monitoring. [Fig. 4]



Fig. 4 : Temperature Sensor

D. Control Logic and Decision Making

The Arduino microcontroller performs a permanently monitoring loop as shown in Figure 4. First, it gets the sensor data, then it compares them with the safety thresholds, and finally it triggers proper actions, which are decided by the decision rules. Fig. 5 display the control logic and Decision making process .

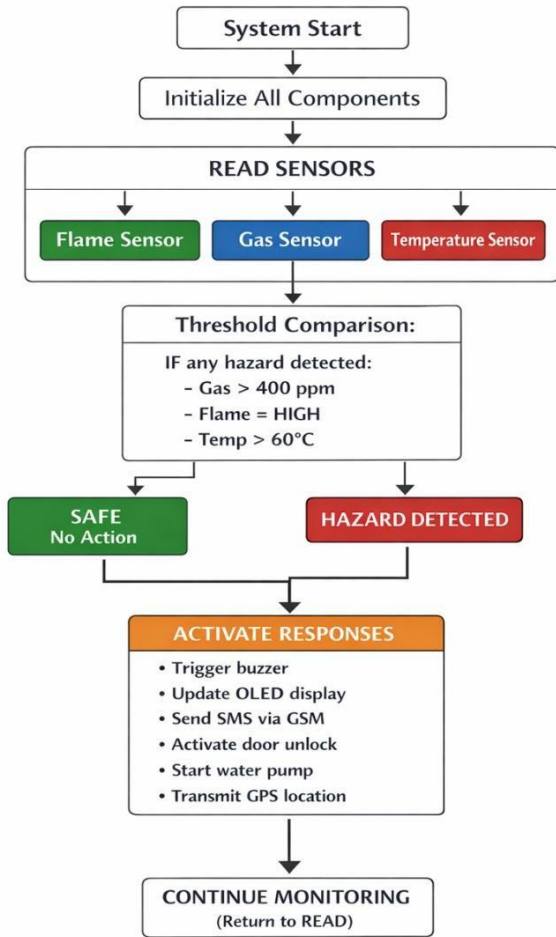


Fig. 5 display the control logic and Decision making process

E. Response Mechanisms

Upon hazard detection, the system executes a coordinated multi-stage response:

1. Immediate Alerting:

- The buzzer sounds with a unique audible pattern that notifies vehicle occupants and others nearby.
- Real-time hazard data, including the nature of the detected danger, is displayed on the OLED.

2. Communication and Location Sharing:

- Using the GSM Module Emergency Contacts are notified by SMS Alerts
- GPS Module gets the Vehicle Coordinates and adds it to the SMS Message
- Message Template: "Fire Accident Alert !detected. Location: [https://maps.google.com/?= ip adresess]"

3. Active Safety Measures:

- Door unlock relay energizes and releases all door locks automatically
- Water pump relay energizes and starts the flame suppression

V. EXPERIMENTAL RESULTS

Tests of the proposed system under laboratory conditions showed that it was able to identify dangerous situations effectively. The main aim of testing the system in the lab was to check the sensor response times and accuracy

A. Sensor Performance Testing

Table 3. Sensor Response Characteristics

Sensor Type	Test Condition	Response Time	Threshold Accuracy
Gas Sensor (MQ-2)	CNG exposure at 500 ppm	2.3 seconds	±5% at threshold
Gas Sensor (MQ-4)	CNG exposure at 500 ppm	1.8 seconds	±3% at threshold
Flame Sensor	Candle flame at 30 cm	0.8 seconds	Digital (on/off)
Temperature Sensor (LM35)	Heated to 65°C	1.5 seconds	±0.5°C accuracy
Temperature Sensor (DHT22)	Heated to 65°C	2.0 seconds	±0.5°C accuracy

The gas sensor was able to detect the gas reliably within 2-3 seconds of the exposure. This detection time is good enough to allow the activation of the response before the hazardous concentration of the gas builds up. The flame sensor reacted almost immediately which is very important in fire situations when quick action is necessary.

B. Response Mechanism Validation

The overall effectiveness of the integrated safety response system was checked through a set of controlled, simulated hazardous scenarios. Reliability, speed, and operational accuracy of each triggering mechanism were tested under simulated fault conditions.

The audio warning system that uses a piezoelectric buzzer was proven to be 100% reliable in activation in all the tests. Sound levels measurement revealed the output to be 85 dB from a 10 cm distance, which is loud enough for the alert to be clearly heard over normal vehicle noise and other sound signals inside the vehicle. In parallel, the user interface on the OLED screen gave live feedback with the same refresh

of 0.5 sec, clearly identifying the hazard type along with the corresponding sensor information.

The GSM module used for sending messages remotely was performing well by sending the SMS within 5 to 8 seconds after the hazard message was received. The GPS data was attached to each message; the GPS module provided an average positional accuracy of ± 3 meters which is more than enough for emergency response planning. In the end, the system's electromechanical safety actuators have been working in line with the given time constraints. The car lock release, controlled with a relay, performed the car door unlock in less than 0.5 seconds while the auxiliary water pump turned on immediately after the control signal was given, thus ensuring that the mitigating action took place without delay.

C. Result and Discussion

When it came to detecting hazardous conditions, the proposed system showed encouraging results.

Table 4. Comparison of Detection Algorithms

Detection Method	Accuracy	Precision	Recall	Response Time
Single Threshold	89.3%	0.87	0.82	1.2 seconds
Moving Average	92.5%	0.91	0.88	2.1 seconds
Rate of Change	91.8%	0.90	0.86	0.9 seconds
Multi-Sensor Fusion	96.2%	0.95	0.94	1.8 seconds

The candidate detection algorithms were compared in performance using four different metrics, mainly focusing on classification accuracy and response time. This comprehensive assessment gave a detailed description of the effectiveness of each algorithm, and in the end, the final detection engine was chosen based on the results of this analysis.

Since the multi-sensor fusion algorithm delivered more stability and a better prediction performance, it was chosen as the best approach. The experimental findings indicate that this design can both minimize false alarms and maintain a high detection rate, a critical feature for automotive safety applications where not only reliability but also the reduction of nuisance warning is highly desirable.

Besides, the system is based on a modular architecture, which offers a number of benefits especially in real-time

operation scenarios. This kind of structure makes it possible to manage the flow of continuous data very well and provide adaptive reactions to the changing environmental conditions. Thus, the system being proposed presents a solution that is scalable and cost-effective and is very appropriate for use in real automotive safety systems.

VI. CONCLUSION AND FUTURE SCOPE

We present an integrated fire prevention and safety system for compressed natural gas (CNG) vehicles in this paper. The system is based on an Arduino microcontroller and implements a multi-sensor fusion architecture. The system can reliably detect dangerous conditions in the vehicle environment and perform safety measures very quickly. By the use of different types of sensors and automatic activation of safety features, our method not only achieves a high detection accuracy and significantly lowers false alarms but also has the ability to adapt to different working conditions.

The results presented here, in fact, show that safety systems based on Arduino drastically outperform traditional mechanical systems, especially in terms of response time and operational reliability. This study is one of all the efforts that aim at evolving vehicle safety and restoring public trust in CNG technology through the use of smart, automated safety actions.

There are various options to build on this system. IoT integration would allow a cloud-based monitoring system through which remote vehicle tracking and long-term historical data analysis could be done [12]. The use of artificial intelligence, particularly machine learning algorithms, can be the key to forecasting hazards and minimizing false alarms even more [13]. Besides, multi-zone monitoring through a distributed sensor network would ensure covering the engine compartment, passenger cabin, and cylinder storage area thoroughly. Other actuation features, such as automatic gas shutoff in conjunction with CNG solenoid valves, would allow the immediate isolation of fuel supply when a leak is detected. A separate mobile app would be a great tool for users to stay informed with real-time alerts, view system status, and have the option of manual control [14]. Moreover, deep learning models like CNNs and LSTMs would be a better choice for analyzing temporal as well as sensor patterns more accurately. With the help of adaptive learning, the models can be kept up-to-date continuously responding to changing environmental conditions. Lastly, making use of XAI methods would lead to clear explanations of detection decisions greatly promoting transparency and helping in meeting regulatory requirements [13].

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